Competitive Exam Practice 6 — Answer Key

Hardwired Control Unit Design

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Q1 (B). Microprogrammed control stores control information in a control memory, so adding a new instruction means adding new microinstructions — a software-like change, hence "facilitates easy implementation of new instructions." It is *slower* per step than hardwired (a memory read per microinstruction), so (A) is false; (C) and (D) are not the distinguishing properties.

Q2 (C). Every hardwired control signal is a Boolean function of an opcode line D_i *and* a step line T_j together — neither alone identifies both "which instruction" and "which step."

Q3 5. $\lceil \log_2 25 \rceil = 5$ (4 bits address only 16 codes, insufficient for 25; 5 bits address 32).

Q4 (C). `SC` counts through control-step numbers and the step decoder expands the count into one-hot T_i lines, exactly as a modulo-n counter plus decoder would.

Q5 64. $2^6 = 64$.

Q6 (C). Hardwired control is *faster* per step than microprogrammed control (pure combinational logic, no memory read) — this is an *advantage* of hardwired control, not a disadvantage, making it the correct choice for "NOT a disadvantage."

Q7 (B). The OR-of-products pattern exists precisely because a control signal can be needed by more than one instruction at the same step number; each instruction that needs it contributes its own $(D_i \cdot T_j)$ term to the same shared equation.